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Multi-layer negative pressure incisional wound therapy dressing

Abstract

A dressing for treating tissue may be a composite of dressing layers, including a contact layer, a manifold layer, and an adhesive drape. The manifold layer may include one or more layers of felted open-cell foam in some examples. The manifold layer may be relatively thin to reduce the dressing profile and increase flexibility, which can enable it to conform to difficult geometry and other tissue sites under negative pressure. The dressing may have a length and a width less than the length. The manifold layer may include a population of holes extending at least partially therethrough, wherein the holes may be configured to promote anisotropic contraction of the dressing parallel to its width. The population of holes may have a circular, ovoid, triangular, square, hexagonal, irregular, or morphous shape. The dressing may be a bolster that may anisotropically contract to provide a closing force to a linear wound.

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Background/Summary

CROSS-REFERENCE TO RELATED APPLICATIONS (1) This application claims the benefit of priority to U.S. Provisional Application No. 62/944,031, filed on Dec. 5, 2019, which is incorporated herein by reference in its entirety.

TECHNICAL FIELD
(1) The invention set forth in the appended claims relates generally to tissue treatment systems and more particularly, but without limitation, to dressings for tissue treatment and methods of using the dressings for tissue treatment.
BACKGROUND
(2) Clinical studies and practice have shown that reducing pressure in proximity to a tissue site can augment and accelerate growth of new tissue at the tissue site. The applications of this phenomenon are numerous, but it has proven particularly advantageous for treating wounds. Regardless of the etiology of a wound, whether trauma, surgery, or another cause, proper care of the wound is important to the outcome. Treatment of wounds or other tissue with reduced pressure may be commonly referred to as “negative-pressure therapy,” but is also known by other names, including “negative-pressure wound therapy,” “reduced-pressure therapy,” “vacuum therapy,” “vacuum-assisted closure,” and “topical negative-pressure,” for example. Negative-pressure therapy may provide a number of benefits, including migration of epithelial and subcutaneous tissues, improved blood flow, and micro-deformation of tissue at a wound site. Together, these benefits can increase development of granulation tissue and reduce healing times.
(3) There is also widespread acceptance that cleansing a tissue site can be highly beneficial for new tissue growth. For example, a wound or a cavity can be washed out with a liquid solution for therapeutic purposes. These practices are commonly referred to as “irrigation” and “lavage” respectively. “Instillation” is another practice that generally refers to a process of slowly introducing fluid to a tissue site and leaving the fluid for a prescribed period of time before removing the fluid. For example, instillation of topical treatment solutions over a wound bed can be combined with negative-pressure therapy to further promote wound healing by loosening soluble contaminants in a wound bed and removing infectious material. As a result, soluble bacterial burden can be decreased, contaminants removed, and the wound cleansed.
(4) While the clinical benefits of negative-pressure therapy and/or instillation therapy are widely known, improvements to therapy systems, components, and processes may benefit healthcare providers and patients.
BRIEF SUMMARY
(5) New and useful systems, apparatuses, and methods for treating tissue in a negative-pressure therapy environment are set forth in the appended claims. Illustrative embodiments are also provided to enable a person skilled in the art to make and use the claimed subject matter.

(6) For example, in some embodiments, a dressing for treating tissue may be a composite of dressing layers, including a contact layer, a manifold layer, and an adhesive drape. The contact layer may include a perforated polymer film in some embodiments. The manifold layer may be multi-layered. The manifold layer may include one or more layers of felted open-cell foam in some examples. The manifold layer may be relatively thin to reduce the dressing profile and increase flexibility, which can enable it to conform to difficult geometry and other tissue sites under negative pressure. The dressing may have a length and a width, wherein the width is less than the length. The manifold layer may include a population of holes extending at least partially through the manifold, wherein the holes may be configured to promote anisotropic contraction of the dressing parallel to the width of the dressing. The population of holes may have a circular, oval, triangular, square, hexagonal, irregular, or amorphous shape. The dressing may be a bolster that may anisotropically contract to provide a closing force to a linear wound.
(7) In some embodiments, the manifold layer may be a felted foam having a firmness factor in a range of about 3 to about 5. In some embodiments, the manifold layer may have a thickness in a range of about 3 millimeters to about 9 millimeters. For example, the manifold layer may have a thickness of about 6 millimeters.
(8) The manifold may be adhered to the polymer film in some embodiments. Suitable bonds between the manifold and the polymer film may include pressure-sensitive adhesive (reactive and non-reactive types); hot melt adhesive (spray applied or deployed as a film, woven, or non-woven); hot press lamination; or flame lamination. The polymer film may also be co-extruded with a bonding layer in-situ, which may be formed from a hot melt adhesive, for example. The dressing may be cut to a desired size before applying the dressing to a tissue. Drape strips or other adhesive strips may be used to seal edges of the dressing and fix the dressing to a patient's skin.
(9) Some embodiments of a dressing for treating a tissue site may include a manifold layer and a contact layer. The manifold layer may include a first side configured to face the tissue site, a second side opposite the first side, and a thickness between the first side and the second side. The manifold layer may further include a plurality of holes extending into the manifold layer on the second side and having a hole depth measured from the second side. The contact layer may be configured to be positioned between the manifold layer and the tissue site. The contact layer may include a plurality of perforations disposed through opposing surfaces of the contact layer.
(10) Some embodiments of a dressing for treating a tissue site with negative pressure may include a contact layer, a first manifold layer coupled to the contact layer, and a second manifold layer coupled to the first manifold layer. The contact layer may be a polymer film including a plurality of perforations through the polymer film. The plurality of perforations may be configured to open. The first manifold layer may be foam having a first density. The second manifold layer may be foam having a second density that is greater than the first density. The second density may be between about 2.6 to about 8.0 lb/ft.sup.3. The second manifold layer may include a first surface and a second surface and a plurality of holes extending therebetween. The plurality of holes may be configured to collapse from a relaxed position to a contracted position in response to an application of negative pressure to the dressing.
(11) Some embodiments of a dressing for treating a tissue site with negative pressure may include a contact layer, a first manifold coupled to the contact layer, and a second manifold layer coupled to the first manifold layer. The contact layer may be a polymer film having a plurality of perforations through the polymer film. The first manifold layer may be foam. The second manifold layer be foam having about 120 to about 250 pores per inch, a plurality of holes through the second manifold layer, and a relaxed length and a relaxed width. The relaxed width may be less than the relaxed length. The second manifold layer may be configured to contract to a contracted width in response to an application of negative pressure to the second manifold layer, wherein the contracted width is less than the relaxed width.
(12) Other embodiments of a dressing for treating a tissue site with negative pressure may include a

contact layer configured to contact the tissue site, a first manifold coupled to the contact layer, and a second manifold coupled to the first manifold opposite the contact layer. The first manifold may be foam. The second manifold may be foam having a plurality of first regions having a first density and a plurality of second regions having a second density less than the first density.

(13) Other embodiments of a dressing for treating a tissue site with negative pressure may include a tissue interface having a contact layer, a first layer coupled to the contact layer, and a second layer coupled to the first layer. The contact layer may be a polymer film having a plurality of perforations through the polymer film. The second layer may include a plurality of holes through the second layer. The tissue interface may have a first length and a first width, wherein the first width is less than the first length. The tissue interface may be configured to contract to a second width in response to an application of negative pressure to the tissue interface, wherein the second width is less than the first width.

(14) A method for treating a tissue site with negative pressure is also described herein, wherein some example embodiments include applying a tissue interface to the tissue site, wherein the tissue interface may include a contact layer, a first manifold layer coupled to the contact layer, and a second manifold layer coupled to the first manifold layer. The contact layer may be a polymer film having a plurality of perforations through the polymer film. The second manifold layer may include a plurality of holes through the second manifold layer. The tissue interface may have a first length and a first width, and the first width may be less than the first length. The tissue interface may be covered with a cover to form a sealed space containing the tissue interface. A fluid conductor may be fluidly coupling to the tissue interface and to a negative-pressure source. Negative pressure from the negative-pressure source may be applied to the tissue interface through fluid conductor. The tissue interface may contract to a second width in response to an application of negative pressure to the tissue interface, wherein the second width is less than the first width.

(15) Other objectives, advantages, and a preferred mode of making and using the claimed subject matter may be understood best by reference to the accompanying drawings in conjunction with the following detailed description of illustrative embodiments.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) FIG. **1** is a block diagram of an example embodiment of a therapy system that can provide negative-pressure treatment and instillation treatment in accordance with this specification;

(2) FIG. **2**A is an exploded view of an example of a tissue interface that can be associated with some embodiments of the therapy system of FIG. **1**;

(3) FIG. **2**B is a cross-sectional view of the assembled example tissue interface of FIG. **2**A along line **2**B-**2**B;

(4) FIG. **3**A is a top view of the tissue interface of FIG. **2**A;

(5) FIG. **3**B is a detail view of the tissue interface of FIG. **3**A taken at reference FIG. **3**B in FIG. **3**A;

(6) FIG. **4** is a top view of an example of a hole having an ovoid shape that may be associated with the tissue interface of FIG. **3**A;

(7) FIG. **5**A is a top view of another example of a tissue interface that can be associated with some embodiments of the therapy system of FIG. **1**;

(8) FIG. **5**B is a detail view of the tissue interface of FIG. **5**A taken at reference FIG. **5**B in FIG. **5**A;

(9) FIG. **6** is a top view of an example of a hole having a hexagonal shape that may be associated with the tissue interface of FIG. **5**A;

(10) FIG. **7**A is a top view of another example of a tissue interface that can be associated with some

embodiments of the therapy system of FIG. **1**;
(11) FIG. **7**B is a detail view of the tissue interface of FIG. **7**A taken at reference FIG. **7**B in FIG. **7**A;
(12) FIG. **8** is a top view of an example of a hole having a circular shape that may be associated with the tissue interface of FIG. **7**A;
(13) FIG. **9**A is a top view of another example of a tissue interface that can be associated with some embodiments of the therapy system of FIG. **1**;
(14) FIG. **9**B is a detail view of the tissue interface of FIG. **9**A taken at reference FIG. **9**B in FIG. **9**A;
(15) FIG. **10** is a top view of an example of a hole having a triangular shape that may be associated with the tissue interface of FIG. **9**A;
(16) FIG. **11**A is an exploded view of another example of a tissue interface that can be associated with some embodiments of the therapy system of FIG. **1**;
(17) FIG. **11**B is a cross-sectional view of the assembled tissue interface of FIG. **11**A along line **11**B-**11**B;
(18) FIG. **12** is an exploded view of another example of a tissue interface that can be associated with some embodiments of the therapy system of FIG. **1**;
(19) FIG. **13**A is a bottom view of an example of a tissue interface that can be associated with some embodiments of the therapy system of FIG. **1**;
(20) FIG. **13**B is a detail view of the tissue interface taken at reference FIG. **13**B in FIG. **13**A;
(21) FIG. **14** is a bottom view of an example of a tissue interface that can be associated with some embodiments of the therapy system of FIG. **1**;
(22) FIG. **15** is an isometric view, with a portion shown in cross-section, of a linear wound;
(23) FIG. **16** is an isometric view, with a portion shown in cross-section, of a portion of an example embodiment of a therapy system being deployed over a linear wound;
(24) FIG. **17** is a top view of an example of a dressing under negative pressure;
(25) FIG. **18**A is a bottom view of an example of the tissue interface of the dressing of FIG. **17**; and
(26) FIG. **18**B is a detail view of the tissue interface taken at reference FIG. **18**B in FIG. **18**A.

DESCRIPTION OF EXAMPLE EMBODIMENTS

(27) The following description of example embodiments provides information that enables a person skilled in the art to make and use the subject matter set forth in the appended claims, but it may omit certain details already well-known in the art. The following detailed description is, therefore, to be taken as illustrative and not limiting.
(28) FIG. **1** is a block diagram of an example embodiment of a therapy system **100** that can provide negative-pressure therapy with instillation of topical treatment solutions to a tissue site in accordance with this specification.
(29) The term "tissue site" in this context broadly refers to a wound, defect, or other treatment target located on or within tissue, including but not limited to, a surface wound, bone tissue, adipose tissue, muscle tissue, neural tissue, dermal tissue, vascular tissue, connective tissue, cartilage, tendons, or ligaments. The term "tissue site" may also refer to areas of any tissue that are not necessarily wounded or defective, but are instead areas in which it may be desirable to add or promote the growth of additional tissue. For example, negative pressure may be applied to a tissue site to grow additional tissue that may be harvested and transplanted. A surface wound, as used herein, is a wound on a body that is exposed to the external environment, such as an injury or damage to the epidermis, dermis, and/or subcutaneous layers. Surface wounds may include ulcers or closed incisions, for example. A surface wound, as used herein, does not include wounds within an intra-abdominal cavity. A wound may include chronic, acute, traumatic, subacute, and dehisced wounds, partial-thickness burns, ulcers (such as diabetic, pressure, or venous insufficiency ulcers), flaps, and grafts, for example.
(30) The therapy system **100** may include a source or supply of negative pressure, such as a

negative-pressure source **105**, and one or more distribution components. A distribution component is preferably detachable and may be disposable, reusable, or recyclable. A dressing, such as a dressing **110**, and a fluid container, such as a container **115**, are examples of distribution components that may be associated with some examples of the therapy system **100**. As illustrated in the example of FIG. **1**, the dressing **110** may include a tissue interface **120**, a cover **125**, or both in some embodiments.

(31) A fluid conductor is another illustrative example of a distribution component. A "fluid conductor," in this context, broadly includes a tube, pipe, hose, conduit, or other structure with one or more lumina or open pathways adapted to convey a fluid between two ends. Typically, a tube is an elongated, cylindrical structure with some flexibility, but the geometry and rigidity may vary. Moreover, some fluid conductors may be molded into or otherwise integrally combined with other components. Distribution components may also include or comprise interfaces or fluid ports to facilitate coupling and de-coupling other components. In some embodiments, for example, a dressing interface may facilitate coupling a fluid conductor to the dressing **110**. For example, such a dressing interface may be a SENSAT.R.A.C.™ Pad available from Kinetic Concepts, Inc. of San Antonio, Texas.

(32) The therapy system **100** may also include a regulator or controller, such as a controller **130**. Additionally, the therapy system **100** may include sensors to measure operating parameters and provide feedback signals to the controller **130** indicative of the operating parameters. As illustrated in FIG. **1**, for example, the therapy system **100** may include a first sensor **135** and a second sensor **140** coupled to the controller **130**.

(33) The therapy system **100** may also include a source of instillation solution. For example, a solution source **145** may be fluidly coupled to the dressing **110**, as illustrated in the example embodiment of FIG. **1**. The solution source **145** may be fluidly coupled to a positive-pressure source, such as a positive-pressure source **150**, a negative-pressure source such as the negative-pressure source **105**, or both in some embodiments. A regulator, such as an instillation regulator **155**, may also be fluidly coupled to the solution source **145** and the dressing **110** to ensure proper dosage of instillation solution (e.g. saline) to a tissue site. For example, the instillation regulator **155** may comprise a piston that can be pneumatically actuated by the negative-pressure source **105** to draw instillation solution from the solution source during a negative-pressure interval and to instill the solution to a dressing during a venting interval. Additionally or alternatively, the controller **130** may be coupled to the negative-pressure source **105**, the positive-pressure source **150**, or both, to control dosage of instillation solution to a tissue site. In some embodiments, the instillation regulator **155** may also be fluidly coupled to the negative-pressure source **105** through the dressing **110**, as illustrated in the example of FIG. **1**.

(34) Some components of the therapy system **100** may be housed within or used in conjunction with other components, such as sensors, processing units, alarm indicators, memory, databases, software, display devices, or user interfaces that further facilitate therapy. For example, in some embodiments, the negative-pressure source **105** may be combined with the controller **130**, the solution source **145**, and other components into a therapy unit.

(35) In general, components of the therapy system **100** may be coupled directly or indirectly. For example, the negative-pressure source **105** may be directly coupled to the container **115** and may be indirectly coupled to the dressing **110** through the container **115**. Coupling may include fluid, mechanical, thermal, electrical, or chemical coupling (such as a chemical bond), or some combination of coupling in some contexts. For example, the negative-pressure source **105** may be electrically coupled to the controller **130** and may be fluidly coupled to one or more distribution components to provide a fluid path to a tissue site. In some embodiments, components may also be coupled by virtue of physical proximity, being integral to a single structure, or being formed from the same piece of material.

(36) A negative-pressure supply, such as the negative-pressure source **105**, may be a reservoir of air

at a negative pressure or may be a manual or electrically-powered device, such as a vacuum pump, a suction pump, a wall suction port available at many healthcare facilities, or a micro-pump, for example. “Negative pressure” generally refers to a pressure less than a local ambient pressure, such as the ambient pressure in a local environment external to a sealed therapeutic environment. In many cases, the local ambient pressure may also be the atmospheric pressure at which a tissue site is located. Alternatively, the pressure may be less than a hydrostatic pressure associated with tissue at the tissue site. Unless otherwise indicated, values of pressure stated herein are gauge pressures. References to increases in negative pressure typically refer to a decrease in absolute pressure, while decreases in negative pressure typically refer to an increase in absolute pressure. While the amount and nature of negative pressure provided by the negative-pressure source **105** may vary according to therapeutic requirements, the pressure is generally a low vacuum, also commonly referred to as a rough vacuum, between −5 mm Hg (−667 Pa) and −500 mm Hg (−66.7 kPa). Common therapeutic ranges are between −50 mm Hg (−6.7 kPa) and −300 mm Hg (−39.9 kPa).
(37) The container **115** is representative of a container, canister, pouch, or other storage component, which can be used to manage exudates and other fluids withdrawn from a tissue site. In many environments, a rigid container may be preferred or required for collecting, storing, and disposing of fluids. In other environments, fluids may be properly disposed of without rigid container storage, and a re-usable container could reduce waste and costs associated with negative-pressure therapy.
(38) A controller, such as the controller **130**, may be a microprocessor or computer programmed to operate one or more components of the therapy system **100**, such as the negative-pressure source **105**. In some embodiments, for example, the controller **130** may be a microcontroller, which generally comprises an integrated circuit containing a processor core and a memory programmed to directly or indirectly control one or more operating parameters of the therapy system **100**. Operating parameters may include the power applied to the negative-pressure source **105**, the pressure generated by the negative-pressure source **105**, or the pressure distributed to the tissue interface **120**, for example. The controller **130** is also preferably configured to receive one or more input signals, such as a feedback signal, and programmed to modify one or more operating parameters based on the input signals.
(39) Sensors, such as the first sensor **135** and the second sensor **140**, are generally known in the art as any apparatus operable to detect or measure a physical phenomenon or property, and generally provide a signal indicative of the phenomenon or property that is detected or measured. For example, the first sensor **135** and the second sensor **140** may be configured to measure one or more operating parameters of the therapy system **100**. In some embodiments, the first sensor **135** may be a transducer configured to measure pressure in a pneumatic pathway and convert the measurement to a signal indicative of the pressure measured. In some embodiments, for example, the first sensor **135** may be a piezo-resistive strain gauge. The second sensor **140** may optionally measure operating parameters of the negative-pressure source **105**, such as a voltage or current, in some embodiments. Preferably, the signals from the first sensor **135** and the second sensor **140** are suitable as an input signal to the controller **130**, but some signal conditioning may be appropriate in some embodiments. For example, the signal may need to be filtered or amplified before it can be processed by the controller **130**. Typically, the signal is an electrical signal, but may be represented in other forms, such as an optical signal.
(40) The tissue interface **120** can be generally adapted to partially or fully contact a tissue site. The tissue interface **120** may take many forms, and may have many sizes, shapes, or thicknesses, depending on a variety of factors, such as the type of treatment being implemented or the nature and size of a tissue site. For example, the size and shape of the tissue interface **120** may be adapted to the contours of deep and irregular shaped tissue sites. Any or all of the surfaces of the tissue interface **120** may have an uneven, coarse, or jagged profile.
(41) In some embodiments, the tissue interface **120** may include a manifold. A manifold in this context may include a means for collecting or distributing fluid across the tissue interface **120**

under pressure. For example, a manifold may be adapted to receive negative pressure from a source and distribute negative pressure through multiple apertures across the tissue interface **120**, which may have the effect of collecting fluid from across a tissue site and drawing the fluid toward the source. In some embodiments, the fluid path may be reversed or a secondary fluid path may be provided to facilitate delivering fluid, such as fluid from a source of instillation solution, across a tissue site.

(42) In some embodiments, the cover **125** may provide a bacterial barrier and protection from physical trauma. The cover **125** may also be constructed from a material that can reduce evaporative losses and provide a fluid seal between two components or two environments, such as between a therapeutic environment and a local external environment. The cover **125** may be, for example, an elastomeric film or membrane that can provide a seal adequate to maintain a negative pressure at a tissue site for a given negative-pressure source. The cover **125** may have a high moisture-vapor transmission rate (MVTR) in some applications. For example, the MVTR may be at least 250 grams per square meter per twenty-four hours in some embodiments, measured using an upright cup technique according to ASTM E96/E96M Upright Cup Method at 38° C. and 10% relative humidity (RH). In some embodiments, an MVTR up to 5,000 grams per square meter per twenty-four hours may provide effective breathability and mechanical properties.

(43) In some example embodiments, the cover **125** may be a polymer drape, such as a polyurethane film, that is permeable to water vapor but impermeable to liquid. Such drapes typically have a thickness in the range of 25-50 microns. For permeable materials, the permeability generally should be low enough that a desired negative pressure may be maintained. The cover **125** may be, for example, one or more of the following materials: polyurethane (PU), such as hydrophilic polyurethane; cellulosics; hydrophilic polyamides; polyvinyl alcohol; polyvinyl pyrrolidone; hydrophilic acrylics; silicones, such as hydrophilic silicone elastomers; natural rubbers; polyisoprene; styrene butadiene rubber; chloroprene rubber; polybutadiene; nitrile rubber; butyl rubber; ethylene propylene rubber; ethylene propylene diene monomer; chlorosulfonated polyethylene; polysulfide rubber; ethylene vinyl acetate (EVA); co-polyester; and polyether block polyamide copolymers. Such materials are commercially available as, for example, Tegaderm® drape, commercially available from 3M Company, Minneapolis Minn.; polyurethane (PU) drape; polyether block polyamide copolymer (PEBAX), for example; and INSPIRE 2301 and INSPIRE 2327 polyurethane films, commercially available from Expopack Advanced Coatings, Wrexham, United Kingdom. In some embodiments, the cover **125** may comprise INSPIRE 2301 having an MVTR (upright cup technique) of 2600 g/m.sup.2/24 hours and a thickness of about 30 microns.

(44) An attachment device may be used to attach the cover **125** to an attachment surface, such as undamaged epidermis, a gasket, or another cover. The attachment device may take many forms. For example, an attachment device may be a medically-acceptable, pressure-sensitive adhesive configured to bond the cover **125** to epidermis around a tissue site. In some embodiments, for example, some or all of the cover **125** may be coated with an adhesive, such as an acrylic adhesive, which may have a coating weight of about 25-65 grams per square meter (g.s.m.). Thicker adhesives, or combinations of adhesives, may be applied in some embodiments to improve the seal and reduce leaks. Other example embodiments of an attachment device may include a double-sided tape, paste, hydrocolloid, hydrogel, silicone gel, or organogel.

(45) The solution source **145** may also be representative of a container, canister, pouch, bag, or other storage component, which can provide a solution for instillation therapy. Compositions of solutions may vary according to a prescribed therapy, but examples of solutions that may be suitable for some prescriptions include hypochlorite-based solutions, silver nitrate (0.5%), sulfur-based solutions, biguanides, cationic solutions, and isotonic solutions.

(46) In operation, the tissue interface **120** may be placed within, over, on, or otherwise proximate to a tissue site. If the tissue site is a wound, for example, the tissue interface **120** may partially or completely fill the wound, or it may be placed over the wound. The cover **125** may be placed over

the tissue interface **120** and sealed to an attachment surface near a tissue site. For example, the cover **125** may be sealed to undamaged epidermis peripheral to a tissue site. Thus, the dressing **110** can provide a sealed therapeutic environment proximate to a tissue site, substantially isolated from the external environment, and the negative-pressure source **105** can reduce pressure in the sealed therapeutic environment.

(47) The fluid mechanics of using a negative-pressure source to reduce pressure in another component or location, such as within a sealed therapeutic environment, can be mathematically complex. However, the basic principles of fluid mechanics applicable to negative-pressure therapy and instillation are generally well-known to those skilled in the art, and the process of reducing pressure may be described illustratively herein as "delivering," "distributing," or "generating" negative pressure, for example.

(48) In general, exudate and other fluid flow toward lower pressure along a fluid path. Thus, the term "downstream" may refer to a location in a fluid path relatively closer to a source of negative pressure or further away from a source of positive pressure. Conversely, the term "upstream" may refer to a location further away from a source of negative pressure or closer to a source of positive pressure. Similarly, it may be convenient to describe certain features in terms of fluid "inlet" or "outlet" in such a frame of reference. This orientation is generally presumed for purposes of describing various features and components herein. However, the fluid path may also be reversed in some applications, such as by substituting a positive-pressure source for a negative-pressure source.

(49) Negative pressure applied across the tissue site through the tissue interface **120** in the sealed therapeutic environment can induce macro-strain and micro-strain in the tissue site. Negative pressure can also remove exudate and other fluid from a tissue site, which can be collected in container **115**.

(50) In some embodiments, the controller **130** may receive and process data from one or more sensors, such as the first sensor **135**. The controller **130** may also control the operation of one or more components of the therapy system **100** to manage the pressure delivered to the tissue interface **120**. In some embodiments, controller **130** may include an input for receiving a desired target pressure and may be programmed for processing data relating to the setting and inputting of the target pressure to be applied to the tissue interface **120**. In some example embodiments, the target pressure may be a fixed pressure value set by an operator as the target negative pressure desired for therapy at a tissue site and then provided as input to the controller **130**. The target pressure may vary from tissue site to tissue site based on the type of tissue forming a tissue site, the type of injury or wound (if any), the medical condition of the patient, and the preference of the attending physician. After selecting a desired target pressure, the controller **130** can operate the negative-pressure source **105** in one or more control modes based on the target pressure and may receive feedback from one or more sensors to maintain the target pressure at the tissue interface **120**.

(51) In some embodiments, the controller **130** may have a continuous pressure mode, in which the negative-pressure source **105** is operated to provide a constant target negative pressure for the duration of treatment or until manually deactivated. Additionally or alternatively, the controller may have an intermittent pressure mode. For example, the controller **130** can operate the negative-pressure source **105** to cycle between a target pressure and atmospheric pressure. For example, the target pressure may be set at a value of 135 mmHg for a specified period of time (e.g., 5 min), followed by a specified period of time (e.g., 2 min) of deactivation. The cycle can be repeated by activating the negative-pressure source **105**, which can form a square wave pattern between the target pressure and atmospheric pressure.

(52) In some example embodiments, the increase in negative-pressure from ambient pressure to the target pressure may not be instantaneous. For example, the negative-pressure source **105** and the dressing **110** may have an initial rise time. The initial rise time may vary depending on the type of dressing and therapy equipment being used. For example, some therapy systems may increase

negative pressure at a rate of about 20-30 mmHg/second, and other therapy systems may increase negative pressure at a rate of about 5-10 mmHg/second. If the therapy system **100** is operating in an intermittent mode, the repeating rise time may be a value substantially equal to the initial rise time.

(53) In some example dynamic pressure control modes, the target pressure can vary with time. For example, the target pressure may vary in the form of a triangular waveform, varying between a negative pressure of 50 and 135 mmHg with a rise rate of negative pressure set at a rate of 25 mmHg/min. and a descent rate set at 25 mmHg/min. In other embodiments of the therapy system **100**, the triangular waveform may vary between negative pressure of 25 and 135 mmHg with a rise rate of about 30 mmHg/min and a descent rate set at about 30 mmHg/min.

(54) In some embodiments, the controller **130** may control or determine a variable target pressure in a dynamic pressure mode, and the variable target pressure may vary between a maximum and minimum pressure value that may be set as an input prescribed by an operator as the range of desired negative pressure. The variable target pressure may also be processed and controlled by the controller **130**, which can vary the target pressure according to a predetermined waveform, such as a triangular waveform, a sine waveform, or a saw-tooth waveform. In some embodiments, the waveform may be set by an operator as the predetermined or time-varying negative pressure desired for therapy.

(55) In some embodiments, the controller **130** may receive and process data, such as data related to instillation solution provided to the tissue interface **120**. Such data may include the type of instillation solution prescribed by a clinician, the volume of fluid or solution to be instilled to a tissue site ("fill volume"), and the amount of time prescribed for leaving solution at a tissue site ("dwell time") before applying a negative pressure to the tissue site. The fill volume may be, for example, between 10 and 500 mL, and the dwell time may be between one second to 30 minutes. The controller **130** may also control the operation of one or more components of the therapy system **100** to instill solution. For example, the controller **130** may manage fluid distributed from the solution source **145** to the tissue interface **120**. In some embodiments, fluid may be instilled to a tissue site by applying a negative pressure from the negative-pressure source **105** to reduce the pressure at the tissue site, drawing solution into the tissue interface **120**. In some embodiments, solution may be instilled to a tissue site by applying a positive pressure from the positive-pressure source **150** to move solution from the solution source **145** to the tissue interface **120**. Additionally or alternatively, the solution source **145** may be elevated to a height sufficient to allow gravity to move solution into the tissue interface **120**.

(56) The controller **130** may also control the fluid dynamics of instillation by providing a continuous flow of solution or an intermittent flow of solution. Negative pressure may be applied to provide either continuous flow or intermittent flow of solution. The application of negative pressure may be implemented to provide a continuous pressure mode of operation to achieve a continuous flow rate of instillation solution through the tissue interface **120**, or it may be implemented to provide a dynamic pressure mode of operation to vary the flow rate of instillation solution through the tissue interface **120**. Alternatively, the application of negative pressure may be implemented to provide an intermittent mode of operation to allow instillation solution to dwell at the tissue interface **120**. In an intermittent mode, a specific fill volume and dwell time may be provided depending, for example, on the type of tissue site being treated and the type of dressing being utilized. After or during instillation of solution, negative-pressure treatment may be applied. The controller **130** may be utilized to select a mode of operation and the duration of the negative pressure treatment before commencing another instillation cycle.

(57) FIG. **2**A is an exploded view of an example of the tissue interface **120** of FIG. **1**, illustrating additional details that may be associated with some embodiments in which the tissue interface **120** includes more than one layer. Herein, various examples of the tissue interface **120** are described that may be suitable for use with the dressing **110** and the therapy system **100**. Further, features or elements of the tissue interface **120** described herein may be referred to as part of the therapy

system **100** or the dressing **110** without reference to the tissue interface **120**.

(58) In the example of FIG. **2**A, the tissue interface **120** may include a first layer, such as a contact layer **205**, and a second layer, such as a manifold layer **210**. The manifold layer **210** may include a first manifold layer **215** and a second manifold layer **220**. In some embodiments, the contact layer **205** may be disposed adjacent to the manifold layer **210**. For example, the contact layer **205** and the manifold layer **210** may be stacked so that the contact layer **205** is in contact with the manifold layer **210**. The contact layer **205** may also be heat-bonded or adhered to the manifold layer **210** in some embodiments, for example, using hot melt adhesive. The first manifold layer **215** and the second manifold layer **220** may be stacked so that the first manifold layer **215** is in contact with the second manifold layer **220**. The first manifold layer **215** and the second manifold layer **220** may also be heat-bonded or adhered to the manifold layer **210** in some embodiments. In some embodiments, the contact layer **205** optionally includes a low-tack adhesive, which can be configured to hold the tissue interface **120** in place while the cover **125** is applied. The low-tack adhesive may be continuously coated on the contact layer **205** or applied in a pattern.

(59) The contact layer **205** may include a means for controlling or managing fluid flow. In some embodiments, the contact layer **205** may be a fluid control layer comprising a liquid-impermeable, elastomeric material. For example, the contact layer **205** may be a polymer film, such as a polyurethane film. In some embodiments, the contact layer **205** may be the same material as the cover **125**. The contact layer **205** may also have a smooth or matte surface texture in some embodiments. A glossy or shiny finish finer or equal to a grade B3 according to the SPI (Society of the Plastics Industry) standards may be particularly advantageous for some applications. In some embodiments, variations in surface height may be limited to acceptable tolerances. For example, the surface of the contact layer **205** may have a substantially flat surface, with height variations limited to 0.2 millimeters over a centimeter.

(60) In some embodiments, the contact layer **205** may be hydrophobic. The hydrophobicity of the contact layer **205** may vary, but may have a contact angle with water of at least ninety degrees in some embodiments. In some embodiments the contact layer **205** may have a contact angle with water of no more than 150 degrees. For example, in some embodiments, the contact angle of the contact layer **205** may be in a range of at least 90 degrees to about 120 degrees, or in a range of at least 120 degrees to 150 degrees. Water contact angles can be measured using any standard apparatus. Although manual goniometers can be used to visually approximate contact angles, contact angle measuring instruments can often include an integrated system involving a level stage, liquid dropper such as a syringe, camera, and software designed to calculate contact angles more accurately and precisely, among other things. Non-limiting examples of such integrated systems may include the FTÅ125, FTÅ200, FTÅ2000, and FTÅ4000 systems, all commercially available from First Ten Angstroms, Inc., of Portsmouth, VA, and the DTA25, DTA30, and DTA100 systems, all commercially available from Kruss GmbH of Hamburg, Germany. Unless otherwise specified, water contact angles herein are measured using deionized and distilled water on a level sample surface for a sessile drop added from a height of no more than 5 cm in air at 20-25° C. and 20-50% relative humidity. Contact angles herein represent averages of 5-9 measured values, discarding both the highest and lowest measured values. The hydrophobicity of the contact layer **205** may be further enhanced with a hydrophobic coating of other materials, such as silicones and fluorocarbons, either as coated from a liquid, or plasma coated.

(61) The contact layer **205** may also be suitable for welding to other layers, including the manifold layer **210**. For example, the contact layer **205** may be adapted for welding to polyurethane foams using heat, radio frequency (RF) welding, or other methods to generate heat such as ultrasonic welding. RF welding may be particularly suitable for more polar materials, such as polyurethane, polyamides, polyesters and acrylates. Sacrificial polar interfaces may be used to facilitate RF welding of less polar film materials, such as polyethylene. In some embodiments, the manifold layer **210** may be flame laminated to the manifold layer **210**.

(62) The area density of the contact layer **205** may vary according to a prescribed therapy or application. In some embodiments, an area density of less than 40 grams per square meter may be suitable, and an area density of about 20-30 grams per square meter may be particularly advantageous for some applications.
(63) In some embodiments, for example, the contact layer **205** may be a hydrophobic polymer, such as a polyethylene film. The simple and inert structure of polyethylene can provide a surface that interacts little, if any, with biological tissues and fluids, providing a surface that may encourage the free flow of liquids and low adherence, which can be particularly advantageous for many applications. Other suitable polymeric films include polyurethanes, acrylics, polyolefin (such as cyclic olefin copolymers), polyacetates, polyamides, polyesters, copolyesters, PEBAX block copolymers, thermoplastic elastomers, thermoplastic vulcanizates, polyethers, polyvinyl alcohols, polypropylene, polymethylpentene, polycarbonate, styreneics, silicones, fluoropolymers, and acetates. A thickness between 20 microns and 100 microns may be suitable for many applications. Films may be clear, colored, or printed. More polar films suitable for laminating to a polyethylene film include polyamide, co-polyesters, ionomers, and acrylics. To aid in the bond between a polyethylene and polar film, tie layers may be used, such as ethylene vinyl acetate, or modified polyurethanes. An ethyl methyl acrylate (EMA) film may also have suitable hydrophobic and welding properties for some configurations.
(64) The contact layer **205** may have one or more passages, which can be distributed uniformly or randomly across the contact layer **205**. In some embodiments, the passages may be bi-directional and pressure-responsive. For example, each of the passages generally may be an elastic passage that is normally unstrained to substantially reduce liquid flow, and can expand or open in response to a pressure gradient and/or in response to the contraction of the manifold layer **210**. As illustrated in the example of FIG. **2**A, the passages may be perforations **225** in the contact layer **205**. Perforations **225** may be formed by removing material from the contact layer **205**. For example, perforations **225** may be formed by cutting through the contact layer **205**. In the absence of a pressure gradient across the perforations **225**, the perforations **225** may be sufficiently small to form a seal or fluid restriction, which can substantially reduce or prevent liquid flow. Additionally, or alternatively, one or more of the passages may be or may function as an elastomeric valve that is normally closed when unstrained to substantially prevent liquid flow, and can open in response to a pressure gradient and/or in response to the contraction of the manifold layer **210**. In some examples, the passages may be fenestrations in the contact layer **205**. Generally, fenestrations are a species of perforation, and may also be formed by removing material from the contact layer **205**. The amount of material removed and the resulting dimensions of the fenestrations may be up to an order of magnitude less than perforations.
(65) In some embodiments, the perforations **225** may be formed as slots (or fenestrations formed as slits) in the contact layer **205**. In some examples, the perforations **225** may be linear slots having a length less than 4 millimeters and a width less than 1 millimeter. The length may be at least 2 millimeters, and the width may be at least 0.4 millimeters in some embodiments. A length of about 3 millimeters and a width of about 0.8 millimeters may be particularly suitable for many applications, and a tolerance of about 0.1 millimeter may also be acceptable. Such dimensions and tolerances may be achieved with a laser cutter, for example. Slots of such configurations may function as imperfect elastomeric valves that can substantially reduce liquid flow in a normally closed or resting state. For example, such slots may form a flow restriction without being completely closed or sealed. The slots can expand or open wider in response to a pressure gradient and/or in response to the contraction of the manifold layer **210** to allow increased liquid flow.
(66) The manifold layer **210** generally includes a manifold or a manifold layer, which can provide a means for collecting or distributing fluid across the tissue interface **120** under pressure. For example, the manifold layer **210** may be adapted to receive negative pressure from a source and distribute negative pressure through multiple apertures across the tissue interface **120**, which may

have the effect of collecting fluid from across a tissue site and drawing the fluid toward the source. In some embodiments, the fluid path may be reversed or a secondary fluid path may be provided to facilitate delivering fluid, such as from a source of instillation solution, across the tissue interface **120**.

(67) In some illustrative embodiments, the pathways of the manifold layer **210** may be interconnected to improve distribution or collection of fluids. In some illustrative embodiments, the manifold layer **210** may be a porous material having interconnected fluid pathways. Examples of suitable porous material that comprise or can be adapted to form interconnected fluid pathways (e.g., channels) may include cellular foam, including open-cell foam such as reticulated foam; porous tissue collections; and other porous material such as gauze or felted mat that generally include pores, edges, and/or walls. Liquids, gels, and other foams may also include or be cured to include apertures and fluid pathways. In some embodiments, the manifold layer **210** may additionally or alternatively include projections that form interconnected fluid pathways. For example, the manifold layer **210** may be molded to provide surface projections that define interconnected fluid pathways.

(68) In some embodiments, the manifold layer **210** may be a reticulated foam having pore sizes and free volume that may vary according to needs of a prescribed therapy. For example, a reticulated foam having a free volume of at least 90% may be suitable for many therapy applications, and a foam having an average pore size in a range of 400-600 microns may be particularly suitable for some types of therapy. The tensile strength of the manifold layer **210** may also vary according to needs of a prescribed therapy. For example, the tensile strength of a foam may be increased for instillation of topical treatment solutions. The 25% compression load deflection of the manifold layer **210** may be at least 0.35 pounds per square inch, and the 65% compression load deflection may be at least 0.43 pounds per square inch. In some embodiments, the tensile strength of the manifold layer **210** may be at least 10 pounds per square inch. The manifold layer **210** may have a tear strength of at least 2.5 pounds per inch. In some embodiments, the manifold layer **210** may be a foam comprised of polyols such as polyester or polyether, isocyanate such as toluene diisocyanate, and polymerization modifiers such as amines and tin compounds. In some examples, the manifold layer **210** may be a reticulated polyurethane foam such as used in GRANUFOAM™ Dressing or V.A.C. VERAFLO™ Dressing, both available from KCI of San Antonio, Texas.

(69) In some embodiments, one or both of the first manifold layer **215** and the second manifold layer **220** may be formed by a felting process. Any porous foam suitable for felting may be used, including the example foams mentioned herein, such as GRANUFOAM™ Dressing. Felting comprises a thermoforming process that permanently compresses a foam to increase the density of the foam while maintaining interconnected pathways. Felting may be performed by any known methods, which may include applying heat and pressure to a porous material or foam material. Some methods may include compressing a foam blank between one or more heated platens or dies (not shown) for a specified period of time and at a specified temperature. The direction of compression may be along the thickness of the foam blank.

(70) The period of time of compression may range from 10 minutes up to 24 hours, though the time period may be more or less depending on the specific type of porous material used. Further, in some examples, the temperature may range between 120° C. to 260° C. Generally, the lower the temperature of the platen, the longer a porous material must be held in compression. After the specified time period has elapsed, the pressure and heat will form a felted structure or surface on or through the porous material or a portion of the porous material.

(71) The felting process may alter certain properties of the original material, including pore shape and/or size, elasticity, density, and density distribution. For example, struts that define pores in the foam may be deformed during the felting process, resulting in flattened pore shapes. The deformed struts can also decrease the elasticity of the foam. The density of the foam is generally increased by felting. In some embodiments, contact with hot-press platens in the felting process can also result

in a density gradient in which the density is greater at the surface and the pores size is smaller at the surface. In some embodiments, the felted structure may be comparatively smoother than any unfinished or non-felted surface or portion of the porous material. Further, the pores in the felted structure may be smaller than the pores throughout any unfinished or non-felted surface or portion of the porous material. In some examples, the felted structure may be applied to all surfaces or portions of the porous material. Further, in some examples, the felted structure may extend into or through an entire thickness of the porous material such that the all of the porous material is felted.
(72) A felted foam may be characterized by a firmness factor, which is indicative of the compression of the foam. The firmness factor of a felted foam can be specified as the ratio of original thickness to final thickness. A compressed or felted foam may have a firmness factor greater than 1. The degree of compression may affect the physical properties of the felted foam. For example, felted foam has an increased effective density compared to a foam of the same material that is not felted. The felting process can also affect fluid-to-foam interactions. For example, as the density increases, compressibility or collapse may decrease. Therefore, foams which have different compressibility or collapse may have different firmness factors. In some example embodiments, a firmness factor can range from about 2 to about 10, preferably about 3 to about 5. For example, the firmness factor of the manifold layer **210** felted foam may be about 5 in some embodiments. There is a general linear relationship between firmness level, density, pore size (or pores per inch) and compressibility. For example, foam that is felted to a firmness factor of 3 will show a three-fold density increase and compress to about a third of its original thickness.
(73) In some embodiments, one or more suitable foam blanks (e.g. of pre-felted foam) may be used for forming one or both of the first manifold layer **215** and the second manifold layer **220**. The foam blank(s) may have about 40 to about 50 pores per inch on average, a density of about 1.3 to about 1.6 lb/ft.sup.3, a free volume of about 90% or more, an average pore size in a range of about 400 to about 600 microns, a 25% compression load deflection of at least 0.35 pounds per square inch, and/or a 65% compression load deflection of at least 0.43 pounds per square inch. In some embodiments, the foam blank(s) may have a thickness greater than 10 millimeters, for example 10-35 millimeters, 10-25 millimeters, 10-20 millimeters, or 15-20 millimeters. In some embodiments, the foam blank(s) may be felted to provide denser foam for one or both of the first manifold layer **215** and the second manifold layer **220**. For example, one or more foam blanks may be felted to a firmness factor of 2-10 to form one or more of the first manifold layer **215** and the second manifold layer **220**. In some embodiments, the foam blank may be felted to a firmness factor of 3-7. Some embodiments may felt the foam blank to a firmness factor of 5.
(74) In some embodiments, the first manifold layer **215** may be an unfelted open-cell foam having a free volume of about 90%, a density in a range of about 1.3 to about 1.60 lb/ft.sup.3, about 40 to about 50 pores per inch on average, and/or average pore size of about 400 to about 600 microns. In some embodiments, the second manifold layer **220** may be an open-cell foam having a free volume in a range of about 9% to about 45%, a density in a range of about 2.6 to about 16 lb/ft.sup.3, about 80 to about 500 pores per inch on average as measured in the direction of compression, an average pore size in a range of about 40 to about 300 microns as measured in the direction of compression, and/or a 25% compression load deflection of about 0.7 to about 3.5 pounds per square inch and a 65% compression load deflection of about 0.86 to about 4.3 pounds per square inch, which may be particularly advantageous under negative pressure. In some embodiments, the second manifold layer **220** may be an open-cell foam having a free volume in a range of about 18% to about 45%, a density in a range of about 2.6 to about 8 lb/ft.sup.3, about 80 to about 250 pores per inch on average (e.g., as measured in the direction of compression), an average pore size in a range of about 80 to about 300 microns (e.g., as measured in the direction of compression), and/or a 25% compression load deflection of about 0.7 to about 1.75 pounds per square inch and a 65% compression load deflection of about 0.86 to about 2.15 pounds per square inch, which may be particularly advantageous under negative pressure. For example, the denser second manifold layer

220 may better resist the compressive effects when used under a compression garment and/or may better maintain fluid flow when under negative pressure.

(75) In some embodiments, the density of the second manifold layer **220** may be about 3.9 to about 4.8 lb/ft.sup.3. In some embodiments, the free volume of the second manifold layer **220** may be about 30%. In some embodiments, the average pore size of the second manifold layer **220** may be about 133 to about 200 microns. In some embodiments, the second manifold layer **220** may have about 120 to about 150 pores per inch on average. In some embodiments, the second manifold layer **220** may have a 25% compression load deflection of at least 1.05 pounds per square inch and a 65% compression load deflection of at least 1.29 pounds per square inch. In some embodiments, the density of the second manifold layer **220** may be about 6.5 to about 8.0 lb/ft.sup.3. In some embodiments, the free volume of the second manifold layer **220** may be about 18%. In some embodiments, the average pore size of the second manifold layer **220** may be about 80 to about 120 microns. In some embodiments, the second manifold layer **220** may have about 200 to about 250 pores per inch on average. In some embodiments, the second manifold layer **220** may have a 25% compression load deflection of at least 1.75 pounds per square inch and a 65% compression load deflection of at least 2.15 pounds per square inch. In some embodiments the second manifold layer **220** may have a higher density than the density of the first manifold layer **215**.

(76) In some embodiments, the first manifold layer **215** may be felted. In some embodiments, the first manifold layer **215** may be an open-cell foam having a free volume in a range of about 9% to about 45%, a density in a range of about 2.6 to about 16 lb/ft.sup.3, about 80 to about 500 pores per inch on average as measured in the direction of compression, an average pore size in a range of about 40 to about 300 microns as measured in the direction of compression, and/or a 25% compression load deflection of about 0.7 to about 3.5 pounds per square inch and a 65% compression load deflection of about 0.86 to about 4.3 pounds per square inch, which may be particularly advantageous under negative pressure. In some embodiments, the first manifold layer **215** may be an open-cell foam having a free volume in a range of about 18% to about 45%, a density in a range of about 2.6 to about 8 lb/ft.sup.3, about 80 to about 250 pores per inch on average as measured in the direction of compression, an average pore size in a range of about 80 to about 300 microns as measured in the direction of compression, and/or a 25% compression load deflection of about 0.7 to about 1.75 pounds per square inch and a 65% compression load deflection of about 0.86 to about 2.15 pounds per square inch, which may be particularly advantageous under negative pressure. For example, the denser first manifold layer **215** may better resist the compressive effects when used under a compression garment and/or may better maintain fluid flow when under negative pressure.

(77) In some embodiments, the density of the first manifold layer **215** may be about 3.9 to about 4.8 lb/ft.sup.3. In some embodiments, the free volume of the first manifold layer **215** may be about 30%. In some embodiments, the average pore size of the first manifold layer **215** may be about 133 to about 200 microns. In some embodiments, the first manifold layer **215** may have about 120 to about 150 pores per inch on average. In some embodiments, the first manifold layer **215** may have a 25% compression load deflection of at least 1.05 pounds per square inch and a 65% compression load deflection of at least 1.29 pounds per square inch. In some embodiments, the density of the first manifold layer **215** may be about 6.5 to about 8.0 lb/ft.sup.3. In some embodiments, the free volume of the first manifold layer **215** may be about 18%. In some embodiments, the average pore size of the first manifold layer **215** may be about 80 to about 120 microns. In some embodiments, the first manifold layer **215** may have about 200 to about 250 pores per inch on average. In some embodiments, the first manifold layer **215** may have a 25% compression load deflection of at least 1.75 pounds per square inch and a 65% compression load deflection of at least 2.15 pounds per square inch.

(78) In some embodiments, only the first manifold layer **215** is felted. In some embodiments, only the second manifold layer **220** is felted. In some embodiments, both the first manifold layer **215**

and the second manifold layer **220** are felted. In embodiments in which the first manifold layer **215** is felted, the contact layer **205** may be coupled to the first manifold layer **215** during the felting process.

(79) As further shown in FIG. **2**A, the second manifold layer **220** may have one or more holes **230**, which can be distributed uniformly or randomly across the second manifold layer **220**. The plurality of holes **230** extending through the second manifold layer **220** may form walls **235** extending through the second manifold layer **220**. The holes **230** may be configured to provide anisotropic contractive properties to the tissue interface **120** in response to the application of negative pressure to the tissue interface **120**.

(80) FIG. **2**B is a cross-sectional view of the assembled tissue interface **120** of FIG. **2**A along line **2**B-**2**B. As shown in FIG. **2**B, the first manifold layer **215** may include a first side **240** configured to face a tissue site, a second side **245** opposite the first side **240**, and a thickness T.sub.1 between the first side **240** and the second side **245**. The second manifold layer **220** may include a first side **250** configured to face a tissue site, a second side **255** opposite the first side **250**, and a thickness T.sub.2 between the first side **250** and the second side **255**. The first side **250** of the second manifold layer **220** may be coupled to the second side **245** of the first manifold layer **215**. If coupled together, the first manifold layer **215** and the second manifold layer **220** may form the manifold layer **210**. In some embodiments, the first side **240** of the first manifold layer **215** may be on a first side of the manifold layer **210** and the second side **255** of the second manifold layer **220** may be on a second side of the manifold layer **210**. In some embodiments, the manifold layer **210** may have a thickness T.sub.M equal to the sum of the thickness T.sub.1 of the first manifold layer **215** and the thickness T.sub.2 of the second manifold layer **220**. In some embodiments, for example, the thickness T.sub.1 of the first manifold layer **215** may be in a range of about 0.5 millimeters to about 5 millimeters. In some embodiments, the thickness T.sub.1 of the first manifold layer **215** may be in a range of about 1 millimeter to about 3 millimeters. In some embodiments, the thickness T.sub.1 of the first manifold layer **215** may be about 2 millimeters. In some embodiments, for example, the thickness T.sub.2 of the second manifold layer **220** may be in a range of about 3 millimeters to about 9 millimeters. In some embodiments, for example, the thickness T.sub.2 of the second manifold layer **220** may be in a range of about 3 millimeters to about 6 millimeters. In some embodiments, the thickness T.sub.2 of the second manifold layer **220** may be about 6 millimeters. In some embodiments, the thickness T.sub.M of the manifold layer **210** may be in a range of about 3 millimeters to about 14 millimeters. In some embodiments, the thickness T.sub.M of the manifold layer **210** may be in a range of about 3 millimeters to about 9 millimeters. In some embodiments, the thickness T.sub.M of the manifold layer **210** may be about 8 millimeters.

(81) With continued reference to FIG. **2**B, the plurality of holes **230** may extend into the second manifold layer **220** on the second side **255** and may have a hole depth D.sub.H measured from the second side **255**. In some embodiments, the hold depth D.sub.H may be equal to the thickness T.sub.2 of the second manifold layer **220**. In some embodiments, the walls **235** may be parallel to the thickness T.sub.2 of the second manifold layer **220**. In other embodiments, the walls **235** may be generally perpendicular to the first side **250** and the second side **255** of the second manifold layer **220**.

(82) FIG. **3**A is a top view of the tissue interface **120** of FIG. **2**A. As shown in FIG. **3**A, in some embodiments, the tissue interface **120** may have a first orientation line **305** and a second orientation line **310** that is perpendicular to the first orientation line **305**. In some embodiments, the first orientation line **305** may be parallel to the length L.sub.TI of the tissue interface **120** and the second orientation line **310** may be parallel to the width W.sub.TI of the tissue interface **120**. Generally, the first orientation line **305** and the second orientation line **310** aid in the description of the tissue interface **120**. In some embodiments, the tissue interface **120** may be configured to contract under the application of negative pressure more in a first direction than in a second

direction. The first orientation line **305** and the second orientation line **310** may be used to refer to a desired direction of contraction of the tissue interface **120**. For example, the desired direction of contraction may be parallel to the second orientation line **310** and perpendicular to the first orientation line **305**. In some embodiments, the second orientation line **310** may be a primary contraction axis. In other embodiments, the desired direction of contraction may be parallel to the first orientation line **305** and perpendicular to the second orientation line **310**. In still other embodiments, the desired direction of contraction may be at a non-perpendicular angle to both the first orientation line **305** and the second orientation line **310**.

(83) As further shown in FIG. **3**A, the tissue interface **120** may have a length L.sub.TI and a width W.sub.TI. In some embodiments, if the tissue interface **120** is not subjected to negative pressure, the length L.sub.TI may be a nominal or relaxed length, and the width W.sub.TI may be a nominal or relaxed width. In some embodiments, the length L.sub.TI may be greater than the width W.sub.TI, such that the width W.sub.TI, may be less than the length L.sub.TI. For example, the tissue interface **120** may have an elongate shape. In some embodiments, for example, the tissue interface **120** may have a length L.sub.TI to width W.sub.TI ratio of at least 4:1. In some embodiments, the length L.sub.TI may be equal to the width W.sub.TI. In some embodiments, the length L.sub.TI may be less than the width W.sub.TI. In some embodiments, one or more of the contact layer **205**, the first manifold layer **215**, and the second manifold layer **220** may be coextensive or congruent with one another. For example, the contact layer **205**, the first manifold layer **215**, and the second manifold layer **220** may have the same length and width. In some embodiments, the contact layer **205**, the first manifold layer **215**, and the second manifold layer **220** may have the same size and shape. In some embodiments, the length and width of the contact layer **205**, the first manifold layer **215**, and the second manifold layer **220** may be equal to the length L.sub.TI and a width W.sub.TI of the tissue interface **120**. In some embodiments, the length and width of the manifold layer **210** may be equal to the length L.sub.TI and a width W.sub.TI of the tissue interface **120**. In some embodiments, each of the contact layer **205**, the first manifold layer **215**, and the second manifold layer **220** may have an elongate length. In some embodiments, each of the contact layer **205**, the first manifold layer **215**, and the second manifold layer **220** may have a length and a width, wherein the length is greater than the width.

(84) Although the tissue interface **120** is shown as having an elongate stadium shape, the tissue interface **120** may have other shapes. For example, the tissue interface **120** may have a rectangular, diamond, square, or circular shape. In some embodiments, the shape of the tissue interface **120** may be selected to accommodate the type of tissue site being treated. For example, the tissue interface **120** may have an oval or circular shape to accommodate an oval or circular tissue site. In some embodiments, each of the contact layer **205**, the first manifold layer **215**, and the second manifold layer **220** may have an elongate stadium shape.

(85) As further illustrated in FIG. **3**A, the holes **230** may include a first plurality of holes **315** and a second plurality of holes **320**. Each of the first plurality of holes **315** and the second plurality of holes **320** may be ovoid shaped. In some embodiments where the holes **230** are ovoid shaped, each of the first plurality of holes **315** may have a length L.sub.1 and a width W.sub.1 perpendicular to the length L.sub.1, and each of the second plurality of holes **320** may have a length L.sub.2 and a width W.sub.2 perpendicular to the length L.sub.2. In some embodiments, the length L.sub.1 may be equal to the length L.sub.2. In some embodiments, the width W.sub.1 may be equal to the width W.sub.2. In the example of FIG. **3**A, L.sub.1 and L.sub.2 may be substantially equal, and W.sub.1 and W.sub.2 may be substantially equal, within acceptable manufacturing tolerances. In some embodiments, L.sub.1 and L.sub.2 may be about 10 millimeters, and W.sub.1 and W.sub.2 may be about 5 millimeters.

(86) The first plurality of holes **315** and the second plurality of holes **320** may be distributed across the second manifold layer **220** in one or more rows in one direction or in different directions. In some embodiments, the rows of the first plurality of holes **315** and the second plurality of holes

320 may be offset or staggered. In some embodiments, the length L.sub.1 of the first plurality of holes **315** and the length L.sub.2 of the second plurality of holes **320** may be positioned at an angle relative to the length L.sub.TI of the tissue interface **120**. In some embodiments, the length L.sub.1 of one or more of the first plurality of holes **315** in a first row may point toward the width W.sub.2 of one or more of the second plurality of holes **320** in a second row. In some embodiments, the length L.sub.2 of one or more of the second plurality of holes **320** in a second row may point toward the width W.sub.1 of one or more of the first plurality of holes **315** in a first row. The pattern of FIG. **3**A may be characterized as a herringbone pattern.
(87) In example embodiments, each of the first plurality of holes **315** may have a first long axis. The first long axis may be parallel to the length L.sub.1 of the first plurality of holes **315**. In some embodiments, the first long axis may be parallel to a first reference line **325** running in a first direction. In illustrative examples, each of the second plurality of holes **320** may have a second long axis. The second long axis may be parallel to the length L.sub.2 of the second plurality of holes **320**. In example embodiments, the second long axis may be parallel to a second reference line **330** running in a second direction. In some embodiments, one or both of the first reference line **325** and the second reference line **330** may be defined relative to the first orientation line **305** of the second manifold layer **220**. For example, one or both of the first reference line **325** and the second reference line **330** may be parallel or coincident with the first orientation line **305** of the second manifold layer **220**. In some illustrative embodiments, one or both of the first reference line **325** and the second reference line **330** may be rotated an angle relative to the first orientation line **305** of the second manifold layer **220**. For example, the first reference line **325** may form an angle θ with the first orientation line **305** and the second reference line **330** may form an angle γ with the first orientation line **305**. In example embodiments, an angle α may define the angle between the first reference line **325** and the second reference line **330**. In some embodiments, the angle θ may be less than about 90°. In some embodiments, the angle θ may be in a range from about 30° to about 70°. In some embodiments, the angle θ may be about 45°. In some embodiments, the angle γ may be greater than about 90°. In some embodiments, the angle γ may be in a range from about 110° to about 150°. In some embodiments, the angle γ may be about 135°. In some embodiments, the angle α may be in a range from about 40° to about 120°. In some embodiments, the angle α may be about 90°.
(88) In some embodiments, as the angle θ between first reference line **325** and the first orientation line **305** decreases and the angle γ between the second reference line **330** and the first orientation line **305** increases, such that the first reference line **325** and the second reference line **330** approach becoming parallel to the first orientation line **305**, the compressibility of the tissue interface **120** parallel to the second orientation line **310** may increase. Consequently, if negative pressure is applied to the tissue interface **120**, the tissue interface **120** may contract more in a direction parallel to the second orientation line **310** than in a direction parallel to the first orientation line **305**. By increasing the compressibility of the tissue interface **120** in a direction parallel to the second orientation line **310**, the tissue interface **120** may collapse to apply a closing force F.sub.C to a tissue site, as described in more detail below.
(89) In some example embodiments, the centroid of each of the first plurality of holes **315** within a row may intersect a third reference line **335** running in a third direction. In illustrative embodiments, the centroid of each of the second plurality of holes **320** within a row may intersect a fourth reference line **340** running in a fourth direction. In the example of FIG. **3**A, the third reference line **335** and the fourth reference line **340** may be parallel to the first orientation line **305**.
(90) The pattern of holes **230** may also be characterized by a pitch, which indicates the spacing between corresponding points on holes **230** within a pattern. In example embodiments, pitch may indicate the spacing between the centroids of holes **230** within the pattern. Some patterns may be characterized by a single pitch value, while others may be characterized by at least two pitch values. For example, if the spacing between centroids of the holes **230** is the same in all

orientations, the pitch may be characterized by a single value indicating the spacing between centroids in adjacent rows. In some embodiments, a pattern comprising a first plurality of holes **315** and a second plurality of holes **320** may be characterized by five pitch values, P.sub.1, P.sub.2, P.sub.3, P.sub.4, and P.sub.5. P.sub.1 may be the spacing between the centroids of each of the first plurality of holes **315** in adjacent rows perpendicular to the third reference line **335**. P.sub.2 may be the spacing between the centroids of each of the second plurality of holes **320** in adjacent rows perpendicular to the fourth reference line **340**. P.sub.3 may be the spacing between adjacent centroids of each of the first plurality of holes **315** within each row parallel to the third reference line **335**. P.sub.4 may be the spacing between adjacent centroids of each of the second plurality of holes **320** within each row parallel to the to the fourth reference line **340**. P.sub.5 may be the spacing between the centroids of each of the first plurality of holes **315** and each of the second plurality of holes **320** in adjacent rows. In the example of FIG. **3**A, P.sub.1 and P.sub.2 may be substantially equal, and P.sub.3 and P.sub.4 may be substantially equal, within acceptable manufacturing tolerances. In some embodiments where P.sub.1 equals P.sub.2 (P.sub.1=P.sub.2), then P.sub.5 may be equal to half of P.sub.1 and P.sub.2 (P.sub.5=0.5×P.sub.1=0.5×P.sub.2). In some embodiments, P.sub.1 and P.sub.2 may be about 17.5 millimeters, P.sub.3 and P.sub.4 may be about 8 millimeters, and P5 may be about 8.75 millimeters.

(91) The plurality of holes **230** may form a percent open area the tissue interface **120**. The percent open area of the plurality of holes **230** may be equal to the percentage of the area of the plurality of holes **230** to the total surface area of the first side of the second manifold layer **220** of the tissue interface **120**. In some embodiments, the percent open area may be between about 40% and about 60%. In other embodiments, the percent open area may be about 56%.

(92) FIG. **3**B is a detail view of the tissue interface **120** taken at reference FIG. **3**B in FIG. **3**A. In some patterns, the rows may be offset or staggered. The stagger may be characterized by an orientation of corresponding points in successive rows relative to an edge or other reference line associated with the second manifold layer **220**. In some embodiments, the rows of the first plurality of holes **315** may be staggered with the rows of the second plurality of holes **320**. In some embodiments, the stagger may be characterized by a stagger value S.sub.1, where S.sub.1 may be the spacing between centroids of each of the first plurality of holes **315** and the second plurality of holes **320** in adjacent rows parallel to the to the fourth reference line **340**. In some embodiments, S.sub.1 may be about 2 to about 10 millimeters. In some embodiments, S.sub.1 may be about 3 millimeters.

(93) In some embodiments, the plurality of walls **235** can be considered to have a concertinaed shape. The concertinaed shape may be formed by a plurality of alternating first wall portions **345** and second wall portions **350**. The first wall portions **345** may be oriented at an angle Φ with respect to the first orientation line **305**. In some embodiments, the angle Φ of the first wall portions **345** may be equal to the angle θ. In the example of FIG. **3**B, the first wall portions **345** may be about 45° with respect to the first orientation line **305**. The second wall portions **350** may be oriented at an angle Ψ with respect to the first orientation line **305**. In some embodiments, the angle Ψ of the second wall portions **350** may be equal to the angle γ. In the example of FIG. **3**B, the second wall portions **350** may be about 135° with respect to the first orientation line **305**. In some embodiments, the first wall portions **345** may be parallel to the length L.sub.1 of the first plurality of holes **315**, and the second wall portions **350** may be parallel to the length L.sub.2 of the second plurality of holes **320**. The first wall portions **345** and second wall portions **350** may have an angle β between each wall portion. In some embodiments, β may be equal to 180° minus α (β=180°−α). In some embodiments, β may be about 90°. In some embodiments, as the angle β between each wall portion decreases, the compressibility of the tissue interface **120** parallel to the second orientation line **310** may increase. Consequently, if negative pressure is applied to the tissue interface **120**, the tissue interface **120** may contract more in a direction parallel to the second orientation line **310** than in a direction parallel to the first orientation line **305**. By increasing the

compressibility of the tissue interface **120** in a direction parallel to the second orientation line **310**, the tissue interface **120** may collapse to apply a closing force F.sub.C to a tissue site, as described in more detail below.

(94) FIG. **4** is a top view of an example of a single hole **230** having an ovoid shape that may be associated with the tissue interface of FIG. **3**A. The hole **230** may include a centroid **405** and a perimeter **410**. The hole **230** may have a perforation shape factor (PSF). The perforation shape factor (PSF) may represent an orientation of the hole **230** relative to the first orientation line **305** and the second orientation line **310**. Generally, the perforation shape factor (PSF) is a ratio of ½ a maximum length of the hole **230** that is parallel to the desired direction of contraction to ½ a maximum length of the hole **230** that is perpendicular to the desired direction of contraction. For descriptive purposes, the desired direction of contraction is parallel to the second orientation line **310**. The desired direction of contraction may be indicated by the closing force F.sub.C. For reference, the hole **230** may have an X-axis **415** extending through the centroid **405** parallel to the first orientation line **305**, and a Y-axis **420** extending through the centroid **405** parallel to the second orientation line **310**. In some embodiments, the perforation shape factor (PSF) of the hole **230** may be defined as a ratio of a line segment **425** on the Y-axis **420** extending from the centroid **405** to the perimeter **410** of the hole **230**, to a line segment **430** on the X-axis **415** extending from the centroid **405** to the perimeter **410** of the hole **230**. For example, if a length of the line segment **425** is 2.5 mm and the length of the line segment **430** is 2.5 mm, the perforation shape factor (PSF) would be 2.5/2.5 or about 1. In some embodiments, the holes **230** may have an ovoid shape as shown. In other embodiments, the holes **230** may have a circular, oval, triangular, square, hexagonal, irregular, or amorphous shape.

(95) In some embodiments, the holes **230** may be formed during molding of the second manifold layer **220**. In other embodiments, the holes **230** may be formed by cutting, melting, or vaporizing the second manifold layer **220** after the second manifold layer **220** is formed. For example, the holes **230** may be formed in the second manifold layer **220** by laser cutting the felted foam of the second manifold layer **220**. In some embodiments, the holes **230** may have an effective diameter, wherein the effective diameter of a non-circular area is defined as a diameter of a circular area having the same surface area as the non-circular area. In some embodiments, each hole **230** may have an effective diameter of about 3.5 millimeters. In other embodiments, each hole **230** may have an effective diameter in a range of about 5 millimeters to about 20 millimeters. The effective diameter of the holes **230** should be distinguished from the porosity of the material forming the walls **235** of the second manifold layer **220**. Generally, an effective diameter of the holes **230** is an order of magnitude larger than the effective diameter of the pores of a material forming the second manifold layer **220**. For example, the effective diameter of the holes **230** may be larger than about 1 millimeters, while the walls **235** may be formed from GRANUFOAM™ Dressing material having a pore size less than about 600 microns. In some embodiments, the pores of the walls **235** may not create openings that extend all the way through the material.

(96) FIG. **5**A is a top view of another example of the tissue interface **120**, illustrating additional details that may be associated with some embodiments of the therapy system **100**. As shown in FIG. **5**A, the plurality of holes **230** may be hexagonal shaped. In some embodiments where the holes are hexagonal shaped, each of the holes **230** may have a length L.sub.3 and a width W.sub.3. In some embodiments, L.sub.3 may be about 5 millimeters, and W.sub.3 may be about 6 millimeters. The plurality of holes **230** may be distributed across the second manifold layer **220** in one or more rows in one direction or in different directions. The pattern of FIG. **5**A may be characterized as a hexagonal pattern. The pattern of the plurality of holes **230** may be described with reference to the angle θ between the first orientation line **305** and the first reference line **325**, the third reference line **335**, the pitch P.sub.3, the pitch P.sub.5, and the stagger S1. In the example of FIG. **5**A, the angle θ may be about 60°, the third reference line **335** may be parallel to the first orientation line **305**, P.sub.3 may be about 7 millimeters, P.sub.5 may be about 5.5 millimeters, and

S.sub.1 may be about 3.5 millimeters. In some embodiments, the pattern of holes **230** in FIG. **5** may be characterized as a staggered pattern of hexagonal holes having a triangular pitch T, where T may be in a range from about 5 millimeters to about 20 millimeters. In some embodiments, the T may be about 15 millimeters. In some embodiments, wherein the plurality of holes **230** have a hexagonal shape, the percent open area may be between about 40% and about 60%. In other embodiments, the percent open area may be about 55%.

(97) FIG. **5**B is a detail view of the tissue interface **120** taken at reference FIG. **5**B in FIG. **5**A. As shown in FIG. **5**B, the plurality of hexagonal holes **230** may cooperate to form the plurality of walls **235** that can be considered to have a concertinaed shape. The concertinaed shape may be formed by a plurality of alternating first wall portions **345** and second wall portions **350**, with a third wall portion **505** in between each first wall portion **345** and second wall portion **350**. In some embodiments, Φ may be about 30°, Ψ may be about 150°, and β may be about 60°. As shown in the example of FIG. **5**B, the third wall portion **505** may be perpendicular to the first orientation line **305**.

(98) FIG. **6** is a top view of an example of a single hole **230** having a hexagonal shape that may be associated with the tissue interface of FIG. **5**A. For reference, the hole **230** may have an X-axis **415** extending through the centroid **405** between opposing sides of the hexagon and parallel to the first orientation line **305**, and a Y-axis **420** extending through the centroid **405** between opposing vertices of the hexagon and parallel to the second orientation line **310**. If a length of the line segment **425** on the Y-axis **420** is 2.69 mm and the length of the line segment **430** on the X-axis is 2.5 mm, the perforation shape factor (PSF) would be 2.69/2.5 or about 1.08. In other embodiments, the hole **230** may be oriented relative to the first orientation line **305** and the second orientation line **310** so that the perforation shape factor (PSF) may be about 1.07 or 1.1.

(99) FIG. **7**A is a top view of another example of the tissue interface **120**, illustrating additional details that may be associated with some embodiments of the therapy system **100**. As shown in FIG. **7**A, the plurality of holes **230** may be circular shaped. In some embodiments where the holes are circular shaped, each of the holes **230** may have a diameter D.sub.1. In some embodiments, D.sub.1 may be about 5 millimeters. The plurality of holes **230** may be distributed across the second manifold layer **220** in one or more rows in one direction or in different directions. The pattern of FIG. **7**A may be characterized as a staggered circular pattern. The pattern of the plurality of holes **230** may be described with reference to the angle θ between the first orientation line **305** and the first reference line **325**, the third reference line **335**, the pitch P.sub.3, the pitch P.sub.5, and the stagger S.sub.1. In the example of FIG. **7**A, the angle θ may be about 37°, the third reference line **335** may be parallel to the first orientation line **305**, P.sub.3 may be about 10 millimeters, P.sub.5 may be about 3.75 millimeters, and S.sub.1 may be about 5 millimeters. In some embodiments, wherein the plurality of holes **230** have a circular shape, the percent open area may be between about 40% and about 60%. In other embodiments, the percent open area may be about 54%.

(100) FIG. **7**B is a detail view of the tissue interface **120** taken at reference FIG. **7**B in FIG. **7**A. As shown in FIG. **7**B, the plurality of circular holes **230** may cooperate to form the plurality of walls **235** that can be considered to have a concertinaed shape. The concertinaed shape may be formed by a plurality of alternating first wall portions **345** and second wall portions **350**. In some embodiments, Φ may be about 37°, Ψ may be about 143°, and β may be about 74°.

(101) FIG. **8** is a top view of an example of a single hole **230** having a circular shape that may be associated with the tissue interface of FIG. **7**A. For reference, the hole **230** may have an X-axis **415** extending through the centroid **405** parallel to the first orientation line **305**, and a Y-axis **420** extending through the centroid **405** parallel to the second orientation line **310**. If a length of the line segment **425** on the Y-axis **420** is 2.5 mm and the length of the line segment **430** on the X-axis is 2.5 mm, the perforation shape factor (PSF) would be 2.5/2.5 or about 1.

(102) FIG. **9**A is a top view of an example of the tissue interface **120**, illustrating additional details

that may be associated with some embodiments of the therapy system **100**. As shown in FIG. **9**A, the holes **230** may include a first plurality of holes **315** and a second plurality of holes **320**. Each of the first plurality of holes **315** and the second plurality of holes **320** may be triangular shaped. The first plurality of holes **315** and the second plurality of holes **320** may be distributed across the second manifold layer **220** in one or more rows in one direction or in different directions. Each of the first plurality of holes **315** may have a first base oriented parallel to the first orientation line **305** and each of the second plurality of holes **320** may have a second base oriented parallel to the first orientation line **305**. Each of the first plurality of holes **315** may have a first apex opposite the first base and each of the second plurality of holes **320** may have a second apex opposite the second base, wherein each first apex points opposite each second apex. In some embodiments, the second manifold layer **220** may have a first edge **905** and a second edge **910** opposite the first edge **905**, wherein each first apex points toward the first edge **905** and each second apex points toward the second edge **910**.

(103) As shown in FIG. **9**A, each row of holes **230** may alternate between a hole of the first plurality of holes **315** and a hole of the second plurality of holes **320**. In some embodiments, the rows of holes **230** may be mirrored. For example, a second base of a hole of the second plurality of holes **320** may be proximate a first base of a hole of the first plurality of holes **315** in an adjacent row. Additionally, a first apex of a hole of the first plurality of holes **315** may be adjacent to a second apex of a hole of the second plurality of holes **320** in an adjacent row.

(104) The pattern of the first plurality of holes **315** and the second plurality of holes **320** may be described with reference to the angle θ between the first orientation line **305** and the first reference line **325**, the angle γ between the first orientation line **305** and the second reference line **330**, third reference line **335**, fourth reference line **340**, pitch values, P.sub.1, P.sub.2, P.sub.3, and P.sub.4, and the stagger S.sub.1. In some embodiments, the angle θ may be in a range from about 50° to about 70°, the angle γ may be about 110° to about 130°, the third reference line **335** may be parallel to the first orientation line **305**, the fourth reference line **340** may be parallel to the first orientation line **305**, P.sub.1 may be in a range from about 5 to about 20 millimeters, P.sub.2 may be about in a range from about 5 to about 20 millimeters, P.sub.3 may be in a range from about 10 to about 40 millimeters, P.sub.4 may be in a range from about 5 to about 20 millimeters, and S.sub.1 may be in a range from about zero to about 10 millimeters. In some embodiments, wherein the plurality of holes **230** have a triangular shape, the percent open area may be between about 40% and about 60%. In other embodiments, the percent open area may be about 40%.

(105) FIG. **9**B is a detail view of the tissue interface **120** taken at reference FIG. **9**B in FIG. **9**A. As shown in FIG. **9**B, the plurality of triangular holes **230** may cooperate to form the plurality of walls **235** that can be considered to have a concertinaed shape. The concertinaed shape may be formed by a plurality of alternating first wall portions **345** and second wall portions **350**. In some embodiments, Φ may be about 63°, Ψ may be about 117°, and β may be about 126°.

(106) FIG. **10** is a top view of an example of a single hole **230** having a triangular shape that may be associated with the tissue interface of FIG. **9**A. For reference, the hole **230** may have an X-axis **415** extending through the centroid **405** parallel to the first orientation line **305**, and a Y-axis **420** extending through the centroid **405** parallel to the second orientation line **310**. If a length of the line segment **425** on the Y-axis **420** is 1.1 mm and the length of the line segment **430** on the X-axis is 1.0 mm, the perforation shape factor (PSF) would be 1.1/1.0 or about 1.1.

(107) FIG. **11**A is an exploded view of an example of the tissue interface **120** of FIG. **1**, illustrating additional details that may be associated with some embodiments in which the tissue interface **120** includes more than one layer. In the example of FIG. **11**A, the tissue interface **120** may include the contact layer **205** and the manifold layer **210**, wherein the manifold layer **210** includes the second manifold layer **220**.

(108) FIG. **11**B is a cross-sectional view of the assembled tissue interface **120** of FIG. **11**A along line **11**B-**11**B. As shown in FIG. **11**B, the tissue interface **120** may include the contact layer **205**

and the second manifold layer **220**. In some embodiments, the contact layer **205** may be coupled directly to the second manifold layer **220**. For example, the contact layer **205** may be coupled to the second manifold layer **220** during the felting process. As further shown in the example of FIG. **11**B, the hole depth D.sub.H of the plurality of holes **230** may be less than the thickness T.sub.2 of the second manifold layer **220**. Thus, the hole depth D.sub.H of the plurality of holes **230** extends partially into the thickness T.sub.2 of the second manifold layer **220** from the second side **255**. For example, in some embodiments, hole depth D.sub.H of the plurality of holes **230** may be less than about 95% of the thickness T.sub.2 of the second manifold layer **220**. In some embodiments, hole depth D.sub.H of the plurality of holes **230** may be less than about 75% of the thickness T.sub.2 of the second manifold layer **220**. In some embodiments, hole depth D.sub.H of the plurality of holes **230** may be less than about 50% of the thickness T.sub.2 of the second manifold layer **220**.
(109) FIG. **12** is an exploded view of an example of the tissue interface **120** of FIG. **1**, illustrating additional details that may be associated with some embodiments in which the tissue interface **120** includes more than one layer. In some embodiments, the second manifold layer **220** may include a plurality of first regions **1205** having a first density and a plurality of second regions **1210** having a second density. In some embodiments, the plurality of first regions **1205** and the plurality of second regions **1210** may extend parallel to the length L.sub.TI of the tissue interface **120**. In some embodiments, the plurality of first regions **1205** and the plurality of second regions **1210** may extend parallel to the width W.sub.TI of the tissue interface **120**. In some embodiments, the plurality of first regions **1205** and the plurality of second regions **1210** may extend at an angle with respect to the length L.sub.TI of the tissue interface **120**. In some embodiments, the tissue interface **120** may be oriented on a tissue site, such as a linear wound, so that the plurality of first regions **1205** and the plurality of second regions **1210** may extend parallel to the linear wound. In some embodiments, the first density may be greater than the second density. In some embodiments, the first density may be in a range from about 2 times to about 5 times greater than the second density. In some embodiments, the plurality of first regions **1205** may be formed by felting portions of the second manifold layer **220**. For example, the plurality of first regions **1205** may be a felted foam having a firmness factor of about 5, and the plurality of second regions **1210** may be an unfelted foam. In some embodiments, the plurality of first regions **1205** may have a free volume in a range of about 18% to about 45%, a density in a range of about 2.6 to about 8.0 lb/ft.sup.3, about 80 to about 250 pores per inch on average (e.g., as measured in the direction of compression), an average pore size in a range of about 80 to about 300 microns (e.g., as measured in the direction of compression), and/or a 25% compression load deflection of about 0.7 to about 1.75 pounds per square inch and a 65% compression load deflection of about 0.86 to about 2.15 pounds per square inch. In some embodiments, the plurality of second regions **1210** may have a free volume in a range of about 90%, a density of about 1.3 lb/ft.sup.3, about 40 to about 50 pores per inch on average, an average pore size in a range of about 400 to about 600 microns, and/or a 25% compression load deflection of about 0.35 pounds per square inch and a 65% compression load deflection of about 0.43 pounds per square inch.
(110) Generally, the plurality of second regions **1210** may be more compressible than the plurality of first regions **1205**. If the tissue interface **120** is placed under a negative-pressure, the plurality of second regions **1210** may collapse before the plurality of first regions **1205**. In some embodiments, if the plurality of second regions **1210** collapse, the tissue interface **120** may compress perpendicular to the plurality of second regions **1210**.
(111) FIG. **13**A is a bottom view of an example of the tissue interface **120**, illustrating additional details of the contact layer **205** that may be associated with some embodiments. FIG. **13**B detail view of the tissue interface **120** taken at reference FIG. **13**B in FIG. **13**A. As illustrated in the example of FIG. **13**A and FIG. **13**B, the perforations **225** may each consist essentially of one or more linear slots having a length L.sub.P and a width W.sub.P, wherein the length L.sub.P extends parallel to the width W.sub.TI of the tissue interface **120** and the width W.sub.P extends parallel to

the length L.sub.TI of the tissue interface **120**. As shown in the FIG. **13**, in some embodiments, the length L.sub.P may extend parallel to the second orientation line **310**, wherein the second orientation line **310** may be the primary contraction axis. A length L.sub.P of about 3 millimeters may be suitable for some examples. A width W.sub.P of about 1 millimeter or less may be suitable for some examples. FIG. **13**A additionally illustrates an example of a uniform distribution pattern of the perforations **225**. In FIG. **13**A, the perforations **225** are substantially coextensive with the contact layer **205**, and are distributed across the contact layer **205** in a grid of parallel rows and columns, in which the slots are also mutually parallel to each other. The rows may be spaced a distance D.sub.1, and the perforations **225** within each of the rows may be spaced a distance D.sub.2. For example, a distance D.sub.1 of about 3 millimeters on center and a distance D.sub.2 of about 3 millimeters on center may be suitable for some embodiments. The perforations **225** in adjacent rows may be aligned or offset. For example, adjacent rows may be offset, as illustrated in FIG. **13**A, so that the perforations **225** are aligned in alternating rows separated by a distance D.sub.3. A distance D.sub.3 of about 6 millimeters may be suitable for some examples. The spacing of the perforations **225** may vary in some embodiments to increase the density of the perforations **225** according to therapeutic requirements.

(112) FIG. **14** is a bottom view of another example of the tissue interface **120**, illustrating additional details of the contact layer **205** that may be associated with some embodiments. As shown in FIG. **14**, in some embodiments, the perforations **225** may each consist essentially of one or more linear slots having a length L.sub.P extending parallel to the length L.sub.TI of the tissue interface **120**.

(113) In some embodiments, one or more of the components of the dressing **110** may additionally be treated with an antimicrobial agent. For example, the manifold layer **210** may be a foam, mesh, or non-woven coated with an antimicrobial agent. In some embodiments, the manifold layer **210** may include antimicrobial elements, such as fibers coated with an antimicrobial agent. Additionally or alternatively, some embodiments of the contact layer **205** may be a polymer coated or mixed with an antimicrobial agent. In other examples, the fluid conductor may additionally or alternatively be treated with one or more antimicrobial agents. Suitable antimicrobial agents may include, for example, metallic silver, PHMB, iodine or its complexes and mixes such as povidone iodine, copper metal compounds, chlorhexidine, or some combination of these materials.

(114) Additionally or alternatively, one or more of the components may be coated with a mixture that may include citric acid and collagen, which can reduce bio-films and infections. For example, the manifold layer **210** may be foam coated with such a mixture.

(115) The cover **125**, the contact layer **205**, the manifold layer **210**, or various combinations may be assembled before application or in situ. For example, the contact layer **205** may be laminated to the manifold layer **210** in some embodiments. In some embodiments, one or more layers of the tissue interface **120** may coextensive. For example, the contact layer **205** and the manifold layer **210** may be cut flush with the edge of the cover **125**, exposing the edge of the manifold layer **210**. In other embodiments, the contact layer **205** may overlap the edge of the manifold layer **210**.

(116) Referring now primarily to FIG. **15** and FIG. **16**, presented is another illustrative embodiment of a portion of the therapy system **100**. FIG. **15** and FIG. **16** depict the therapy system **100** assembled in stages at a tissue site, such as a linear wound **1505**. For example, the linear wound **1505** may have an elongated shape, such as an incision having a length substantially greater than its width. An incision may have edges that may be substantially parallel, particularly if the incision is caused by a scalpel, knife, razor, or other sharp blade. Other examples of a linear wound **1505** may include a laceration, a puncture, or other separation of tissue, which may have been caused by trauma, surgery, or degeneration. In some embodiments, a linear wound **1505** may also be an incision in an organ adjacent a fistula. In some embodiments, a linear wound **1505** may be an incision or puncture in otherwise healthy tissue that extends up to 40 cm or more in length. In some embodiments, a linear wound **1505** may also vary in depth. For example, an incision may have a

depth that extends up to 15 cm or more or may be subcutaneous depending on the type of tissue and the cause of the incision. Additionally shown in FIG. **15**, a closure device **1510**, such as, for example, stitches **1515**, close the linear wound **1505**. Other closure devices **1510**, such as epoxy or staples may be utilized to close the linear wound **1505**. In some embodiments, the linear wound **1505** may be a closed incisional wound. The linear wound **1505** may include a portion through an epidermis **1520**, dermis **1525**, and subcutaneous tissue **1530** of a patient.

(117) Referring now to FIG. **16**, after the linear wound **1505** is closed or prepared as described above, the dressing **110** may be disposed proximate to the linear wound **1505**. The geometry and dimensions of the tissue interface **120**, the cover **125**, or both may vary to suit a particular application or anatomy. For example, the dressing **110** may be cut to size for a specific region or anatomical area, such as for amputations. The dressing **110** may be cut without losing pieces of the tissue interface **120** and without separation of the tissue interface **120**.

(118) The tissue interface **120** can be placed over, on, or otherwise proximate to the linear wound **1505**. In the example of FIG. **16**, the contact layer **205** forms an outer surface of the dressing **110**, and can be placed over the tissue site, including the linear wound **1505** and epidermis **1520**. The contact layer **205** may be interposed between the manifold layer **210** and the tissue site, which can prevent direct contact between the manifold layer **210** and the linear wound **1505** and epidermis **1520**. In some embodiments, the first manifold layer **215** can prevent direct contact between the holes **230** in the second manifold layer **220** and the linear wound **1505** and epidermis **1520**. In some embodiments, the tissue interface **120** may be placed over, on, or otherwise proximate to the linear wound such that the first orientation line **305** is oriented substantially parallel to the linear wound **1505**. In some embodiments, the length L.sub.TI of the tissue interface **120** may be oriented substantially parallel to the linear wound **1505**.

(119) In some examples, the dressing **110** may include one or more attachment devices. In some embodiments, one or more of the attachment devices may include an adhesive **1605**. In some examples the adhesive **1605** may be, for example, a medically-acceptable, pressure-sensitive adhesive that extends about a periphery, a portion, or an entire surface of each of the cover **125**. In some embodiments, for example, the adhesive **1605** may be an acrylic adhesive having a coating weight between 25-65 grams per square meter (g.s.m.). Thicker adhesives, or combinations of adhesives, may be applied in some embodiments to improve the seal and reduce leaks. In some embodiments, such a layer of the adhesive **1605** may be continuous or discontinuous. Discontinuities in the adhesive **1605** may be provided by apertures or holes (not shown) in the adhesive **1605**. The apertures or holes in the adhesive **1605** may be formed after application of the adhesive **1605** or by coating the adhesive **1605** in patterns on a carrier layer, such as, for example, a side of the cover **125**. Apertures or holes in the adhesive **1605** may also be sized to enhance the MVTR of the adhesive **1605** in some example embodiments

(120) The adhesive **1605** can be disposed on a bottom side of the cover **125**, and the adhesive **1605** may pressed onto the cover **125** and epidermis **1520** (or other attachment surface) to fix the dressing **110** in position and to seal the tissue interface **120** over the patient. In some embodiments, the adhesive **1605** can be disposed only around edges of the cover **125**.

(121) FIG. **16** also illustrates one example of a fluid conductor **1610** and a dressing interface **1615**. As shown in the example of FIG. **16**, the fluid conductor **1610** may be a flexible tube, which can be fluidly coupled on one end to the dressing interface **1615**. The dressing interface **1615** may be an elbow connector. In some examples, the tissue interface **120** can be applied to the tissue site before the cover **125** is applied over the tissue interface **120**. The cover **125** may include an aperture **1620**, or the aperture **1620** may be cut into the cover **125** before or after positioning the cover **125** over the tissue interface **120**. The position of the aperture **1620** may be off-center or adjacent to an end or edge of the cover **125**. In other examples, the aperture **1620** may be centrally disposed. The dressing interface **1615** can be placed over the aperture **1620** to provide a fluid path between the fluid conductor **1610** and the tissue interface **120**. In other examples, the fluid conductor **1610** may

be inserted directly through the cover **125** into the tissue interface **120**.

(122) If not already configured, the dressing interface **1615** may be disposed over the aperture **1620** and attached to the cover **125**. The fluid conductor **1610** may be fluidly coupled to the dressing interface **1615** and to the negative-pressure source **105**.

(123) Negative pressure from the negative-pressure source **105** can be distributed through the fluid conductor **1610** and the dressing interface **1615** to the tissue interface **120**. The dressing **110** may be a bolster to aid in closing the linear wound **1505**. The tissue interface **120** may contract in response to the application of negative pressure. In some embodiments, the manifold layer **210** of the tissue interface **120** is configured to anisotropically contract. For example, under an applied negative pressure, the manifold layer **210** may contract more perpendicular to the first orientation line **305** than parallel to the first orientation line **305**. The tissue interface **120** and the dressing **110** may contract more in a width-wise direction than in a length-wise direction. The preferential contraction perpendicular to the first orientation line **305** by the manifold layer **210** creates the closing force F.sub.C which may act to pull the epidermis **1520** toward the linear wound **1505** aiding in closing the linear wound **1505**. The closing force F.sub.C may aid in approximating or closing the subcutaneous tissue **1530** proximate the linear wound **1505**.

(124) FIG. **17** is a top view of an example of a dressing **110** under negative pressure, illustrating additional details that may be associated with some embodiments of the therapy system **100**. The dressing **110** may have an elongate shape. The dressing **110** may have a length and a width, wherein the length is greater than the width. As shown in FIG. **17**, if negative pressure is applied to the tissue interface **120**, the plurality of holes **230** may collapse or contract from a relaxed position to a contracted position. The plurality of holes **230** may collapse from the relaxed position to the contracted position perpendicular to the first orientation line **305**. Additionally, if negative pressure is applied to the tissue interface, and the second manifold layer **220** may collapse or contract to a contracted width, wherein the contracted width is less than a relaxed width in the absence of negative pressure. Under an applied negative pressure, the tissue interface **120** may collapse or contract, in the widthwise direction, to a contracted width W.sub.C that is less than the nominal or relaxed width W.sub.TI of the tissue interface **120**. As the tissue interface **120** contracts perpendicular to the first orientation line **305**, the contraction may be applied to the epidermis **1520** via the contact layer **205** and the cover **125**. The contraction can create the closing force F.sub.C. As the tissue interface **120** contracts in the widthwise direction, the dressing **110** may also contract in the widthwise direction.

(125) FIG. **18**A is a bottom view of the tissue interface **120** of the dressing **110** shown in FIG. **17**. As shown in FIG. **18**A, the tissue interface **120** is shown contracted under negative pressure. FIG. **18**B is a detail view of the tissue interface taken at reference FIG. **18**B in FIG. **18**A. The contact layer **205** may be coupled to the manifold layer **210**. Accordingly, contraction of the manifold layer **210** may be transmitted to the contact layer **205**, causing the contact layer **205** to contract. The contraction of the manifold layer **210** and the contact layer **205** perpendicular to the first orientation line **305** may cause the perforations **225** to open. For example, as shown in FIG. **18**B, each perforation **225** may have a first end **1805** and a second end **1810**, wherein under the application of negative pressure, the contraction of the tissue interface **120** can move the first end **1805** toward the second end **1810** resulting in the perforations **225** having an expanded width W.sub.E and a contracted length L.sub.C, wherein the expanded width W.sub.E is greater than the nominal width W.sub.P of the perforation **225** and the contracted length L.sub.C is less than the nominal length L.sub.P of the perforation **225**. As the tissue interface **120** and the dressing **110** contract in the width-wise direction the perforations **225** may open. Opening the perforations **225** can allow liquid movement through the perforations **225** into the manifold layer **210**. For example, the open perforations **225** may remove moisture from the epidermis **1520** around the linear wound **1505**, which may keep the epidermis **1520** cool and/or dry. In some embodiments, negative pressure applied through the tissue interface **120** can also create a negative pressure differential across the

perforations **225** in the contact layer **205**, which may aid in opening or expanding the perforations **225**. For example, in some embodiments in which the perforations **225** may comprise substantially closed fenestrations through the contact layer **205**, a pressure gradient across the fenestrations can strain the adjacent material of the contact layer **205** and increase the dimensions of the fenestrations to allow liquid movement through them, similar to the operation of a duckbill valve.

(126) The contact layer **205** can protect the epidermis **1520** from irritation that could be caused by expansion, contraction, or other movement of the manifold layer **210**. The contact layer **205** can also substantially reduce or prevent exposure of a tissue site to the manifold layer **210**, which can inhibit growth of tissue into the manifold layer **210**. In some embodiments, the contact layer **205** may also

(127) The systems, apparatuses, and methods described herein may provide significant advantages. The plurality of holes **230** in the manifold layer **210** may be blind in that they do not extend all the way to the contact layer **205**. A typical thin film contact layer may be ineffective at preventing the epidermis **1520** from being drawn into the plurality of holes **230**. Accordingly, providing the first manifold layer **215** between the contact layer **205** and the plurality of holes **230** in the second manifold layer **220** and/or providing a plurality of holes **230** that do not extend all the way through the second manifold layer **220**, provides a buffer layer between the epidermis **1520** and the plurality of holes **230**. Because the plurality of holes **230** are blind, the epidermis **1520** proximate the linear wound **1505** is less likely to be drawn toward and/or into the plurality of holes **230**, resulting in a reduced likelihood of the formation of tissue puckering and macro columns, such as nodules in the epidermis **1520**. Puckering of the epidermis **1520** can result in a plurality of bruises and/or blisters having the same pattern as the plurality of holes. The bruises and/or blisters can be unsightly and may result in discomfort. Accordingly, because the plurality of holes **230** do not touch the contact layer **205**, bruises and/or blisters and any discomfort therefrom can be reduced or eliminated.

(128) The felted construction of the first manifold layer **215** and/or the second manifold layer **220** allows for the manifold layer **210** to be reduced in thickness while still providing good manifolding. For example, in some embodiments, the manifold layer **210** may be about 6 millimeters thick. The thin manifold layer **210** may reduce the overall thickness of the dressing **110**. In some embodiments, the dressing **110** may be about 3 to 5 times thinner than other dressings. The thin dressing **110** reduces bulk and may be easy to apply, especially around difficult geometry and anatomy. Some embodiments of the tissue interface **120** may be easily cut to size to fit the tissue site and may be sealed using a single cover **125**. Embodiments of the thin dressing **110** may reduce or eliminate the occurrence of leaks, reducing power usage of the negative-pressure source **105**. This may be especially beneficial for battery-operated negative-pressure sources **105**. Additionally, in some embodiments, smaller negative-pressure sources **105** may be used with the dressing **110**, wherein such smaller negative-pressure sources **105** may be quieter or silent and may be more discrete. The lower profile dressing **110** may therefore be easy to apply around difficult anatomy, ensuring a good seal, while providing high levels of apposition or closing forces, bolstering effect, and a sterile barrier to potential infection.

(129) The thin dressing **110** may also be more comfortable for the patient. The reduced thickness may reduce or eliminate a “tent area” that can occur between the cover **125** and the tissue interface **120**, thereby reducing or eliminating skin irritation and/or hyperpigmentation around the dressing **110**.

(130) Antimicrobial agents in the dressing **110** may extend the usable life of the dressing **110** by reducing or eliminating infection risks that may be associated with extended use, particularly use with infected or highly exuding wounds.

(131) While shown in a few illustrative embodiments, a person having ordinary skill in the art will recognize that the systems, apparatuses, and methods described herein are susceptible to various changes and modifications that fall within the scope of the appended claims. Moreover,

descriptions of various alternatives using terms such as "or" do not require mutual exclusivity unless clearly required by the context, and the indefinite articles "a" or "an" do not limit the subject to a single instance unless clearly required by the context. Components may be also be combined or eliminated in various configurations for purposes of sale, manufacture, assembly, or use. For example, in some configurations the dressing **110**, the container **115**, or both may be separated from other components for manufacture or sale. In other example configurations, the controller **130** may also be manufactured, configured, assembled, or sold independently of other components.
(132) The appended claims set forth novel and inventive aspects of the subject matter described above, but the claims may also encompass additional subject matter not specifically recited in detail. For example, certain features, elements, or aspects may be omitted from the claims if not necessary to distinguish the novel and inventive features from what is already known to a person having ordinary skill in the art. Features, elements, and aspects described in the context of some embodiments may also be omitted, combined, or replaced by alternative features serving the same, equivalent, or similar purpose without departing from the scope of the invention defined by the appended claims.

Claims

1. A dressing for treating a tissue site, comprising; a manifold layer comprising a single layer of foam having a first side configured to face the tissue site, a second side opposite the first side, and a thickness between the first side and the second side, the manifold layer further comprising a plurality of holes extending into the manifold layer on the second side and having a hole depth measured from the second side, wherein the hole depth of the plurality of holes extends partially into the thickness of the manifold layer from the second side; and a contact layer configured to be positioned between the manifold layer and the tissue site and comprising a plurality of perforations disposed through opposing surfaces of the contact layer.
2. The dressing of claim 1, wherein the contact layer comprises a polymer film.
3. The dressing of claim 1, wherein the plurality of perforations have a perforation length and a perforation width that is shorter than and perpendicular to the perforation length, wherein the perforation length has a first end and a second end opposite the first end, and wherein the perforation width is configured to expand when a contracting force moves the first end of the perforation length closer to the second end of the perforation length.
4. The dressing of claim 1, wherein the manifold layer comprises foam having a density between about 2.6 to about 8.0 lb/ft.sup.3 and a free volume in a range of about 18% to about 45%.
5. The dressing of claim 1, wherein the manifold layer has a thickness between about 3 millimeters to about 9 millimeters.
6. The dressing of claim 1, wherein the plurality of holes includes an elongate length greater than a width perpendicular to the elongate length, and wherein the elongate length of the plurality of holes is positioned at an angle relative to a longitudinal length of the manifold layer.
7. The dressing of claim 1, wherein the manifold layer comprises foam having a plurality of first regions having a first density and a plurality of second regions having a second density less than the first density.
8. The dressing of claim 1, wherein the manifold layer and the contact layer are coextensive with one another.
9. A dressing for treating a tissue site with negative pressure, the dressing comprising: a contact layer comprising a polymer film including a plurality of perforations through the polymer film that are configured to open; a first manifold layer coupled to the contact layer, the first manifold layer comprising foam having a first density; and a second manifold layer coupled to the first manifold layer, the second manifold layer comprising foam having a second density that is greater than the first density, the second density between about 2.6 to about 8.0 lb/ft.sup.3, the second manifold

layer having a first surface and a second surface including a plurality of holes extending therebetween, wherein the plurality of holes are configured to collapse from a relaxed position to a contracted position in response to an application of negative pressure to the dressing.

10. The dressing of claim 9, wherein each of the plurality of holes is adapted to collapse from the relaxed position to the contracted position perpendicular to a line of symmetry of the second manifold layer.

11. The dressing of claim 9, wherein the second manifold layer has a thickness of about 3 to about 9 mm.

12. The dressing of claim 9, wherein the foam of the second manifold layer has a 25% compression load deflection in a range of about 1.05 to about 1.75 pounds per square inch and a 65% compression load deflection in a range of about 1.29 to about 2.15 pounds per square inch.

13. The dressing of claim 9, wherein the foam of the second manifold layer comprises felted foam with a firmness factor of 3-5.

14. The dressing of claim 9, wherein each of the plurality of holes includes a perforation shape factor, wherein the perforation shape factor is a ratio of one half a first maximum length of a hole of the plurality of holes that is parallel to a desired direction of contraction to one half a second maximum length of the hole of the plurality of holes that is perpendicular to the desired direction of contraction.

15. The dressing of claim 9, wherein the contact layer is configured to contact the tissue site, the first manifold layer is in contact with the contact layer, and the second manifold layer is in contact with the first manifold layer opposite the contact layer.

16. A dressing for treating a tissue site with negative pressure, the dressing comprising: a contact layer configured to contact the tissue site, the contact layer comprising a polymer film; a first manifold coupled to the contact layer, the first manifold comprising foam; and a second manifold coupled to the first manifold opposite the contact layer, the second manifold comprising foam having a plurality of first regions having a first density and a plurality of second regions having a second density less than the first density; wherein the contact layer is configured to be between the tissue site and the first manifold when the dressing is applied to the tissue site.

17. The dressing of claim 16, wherein the second manifold has a first length and a first width, and wherein the plurality of first regions and the plurality of second regions extend parallel to the first length.

18. The dressing of claim 17, wherein the second manifold is configured to contract to a second width in response to an application of negative pressure to the second manifold, wherein the second width is less than the first width.

19. The dressing of claim 16, wherein the first density is in a range of about 2.6 to about 8.0 lb/ft.sup.3.